

Time Series Analysis

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An offset detection method

Identification of deterministic offsets in GNSS time series:

$$H_0: E(y) = Ax, D(y) = Qy \quad \text{vs.} \quad H_a: E(y) = Ax + A_k x_k, D(y) = Qy$$

Application to offset detection

$$\text{Model 1 } (H_0) : E(y) = Ax$$

versus

$$\text{Model 2 } (H_a) : E(y) = [A : a_j] \begin{bmatrix} x \\ x_j \end{bmatrix} = Ax + a_j x_j.$$

$$a_j = \begin{bmatrix} H_j(t_1) \\ \vdots \\ H_j(t_m) \end{bmatrix}$$

where $H_j(t_i)$ is the Heaviside step function of the form

$$H_j(t_i) = \begin{cases} 0 & t_i < t_j \\ 1 & t_i \geq t_j, \quad i = 1, \dots, m \end{cases} \quad \rightarrow$$

$$a_j = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$$



Least squares harmonic estimation

The test statistic (least squares power spectrum) is

$$T = P(\omega_k) = \hat{e}^T Q_y^{-1} A_k (A_k^T Q_y^{-1} P_A^\perp A_k)^{-1} A_k^T Q_y^{-1} \hat{e}$$

which becomes

$$P(t_j) = \hat{e}_0^T Q_y^{-1} a_j \left(a_j^T Q_y^{-1} P_A^\perp a_j \right)^{-1} a_j^T Q_y^{-1} \hat{e}_0$$

or

$$P(t_j) = \frac{(\hat{e}_0^T Q_y^{-1} a_j)^2}{a_j^T Q_y^{-1} P_A^\perp a_j}, \quad j = 1, \dots, m$$

Having a chi-squared distribution with one degree of freedom

$$P(t_j) \sim \chi^2(1,0)$$

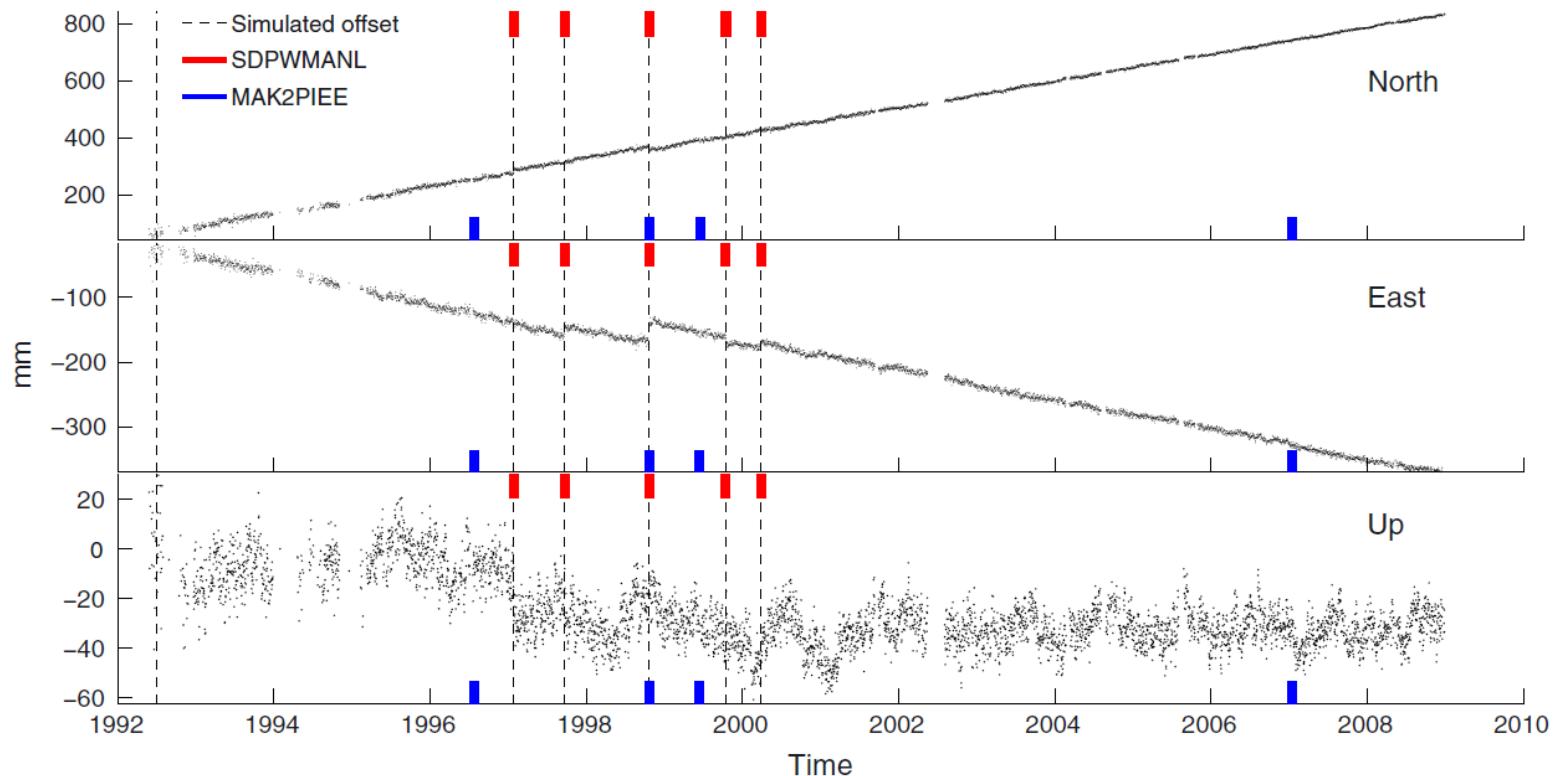


Offset detection

Example I:

Detecting offsets in GPS time series: First results from the detection of offsets in GPS experiment

Julien Gazeaux,¹ Simon Williams,² Matt King,^{1,3} Machiel Bos,⁴ Rolf Dach,⁵
Manoj Deo,⁶ Angelyn W Moore,⁷ Luca Ostini,⁵ Elizabeth Petrie,¹ Marco Roggero,⁸
Felix Norman Teferle,⁹ German Olivares,⁹ and Frank H. Webb⁷

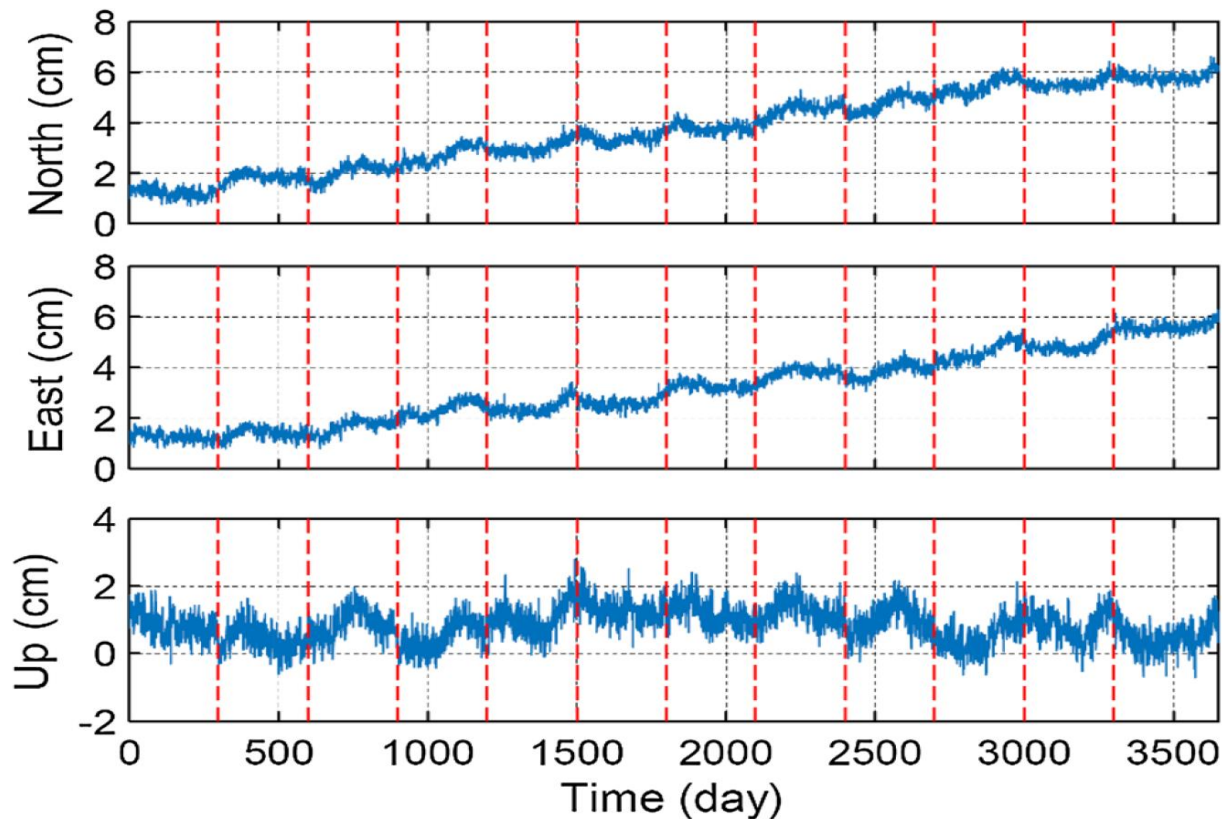


Offset detection

Example II:

Offset detection in GPS position time series using multivariate analysis

A. R. Amiri-Simkooei^{1,2} · M. Hosseini-Asl¹ · J. Asgari¹ · F. Zangeneh-Nejad¹



Offset detection

Simulated data

Table 1 Parameter settings of simulated data sets employed in functional and stochastic models

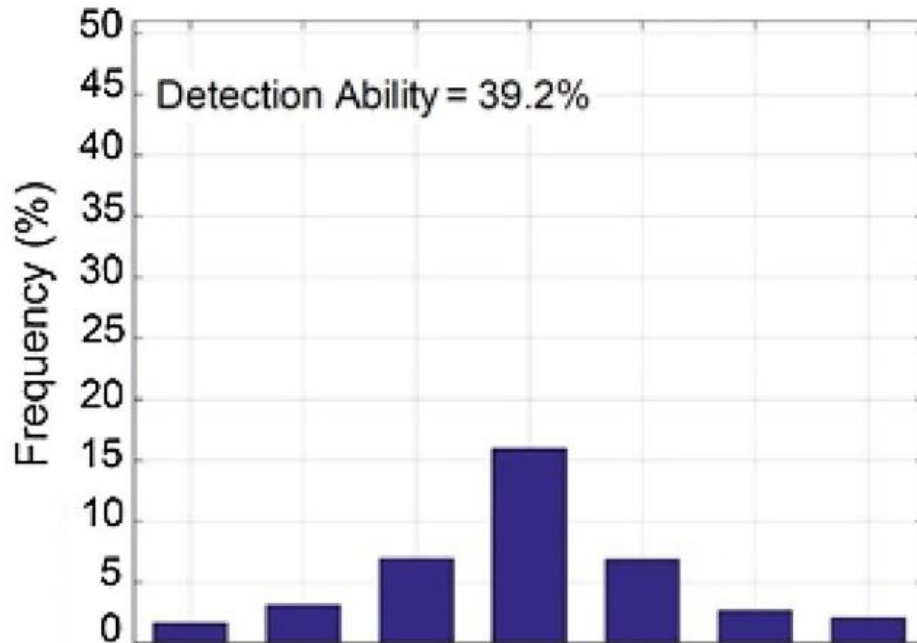
Component	North	East	Up
Parameter			
WN amplitude (mm)	1.5	1.5	3
FN amplitude (mm/year ^{1/4})	3	3	6
Annual amplitude (mm)	2	2	3
Semi-annual amplitude (mm)	1	1	2
v (mm/year)	5	5	1
y_0 (mm)	10	10	10
Offset magnitude (mm)	[1–3]	[1–3]	[2–6]



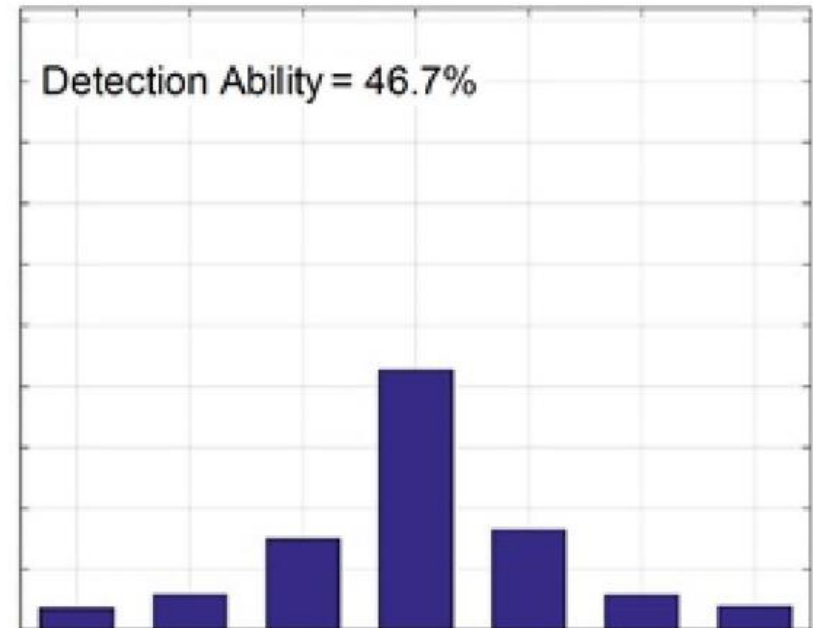
Offset detection power

Univariate analysis: **NORTH**

White noise only



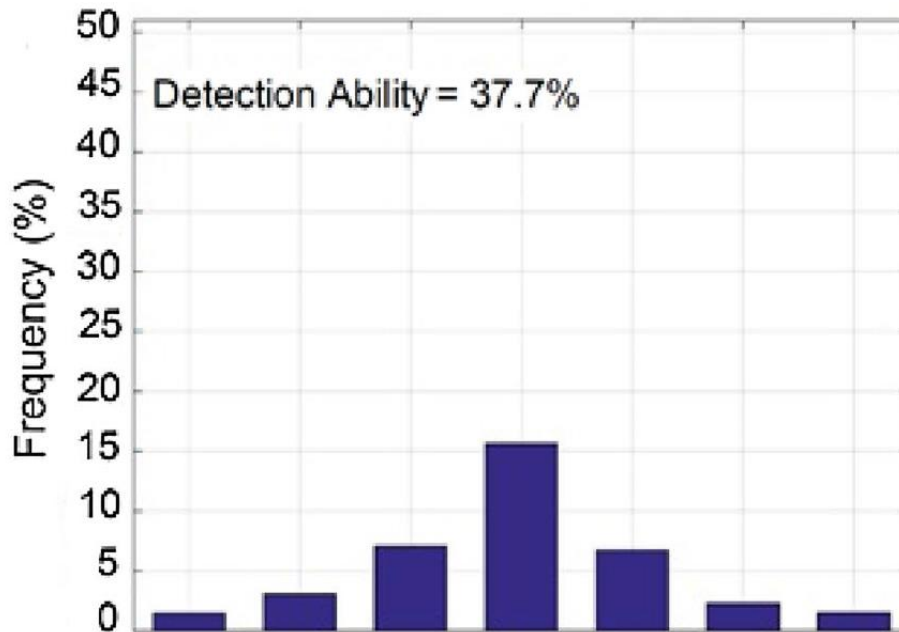
White noise + Flicker noise



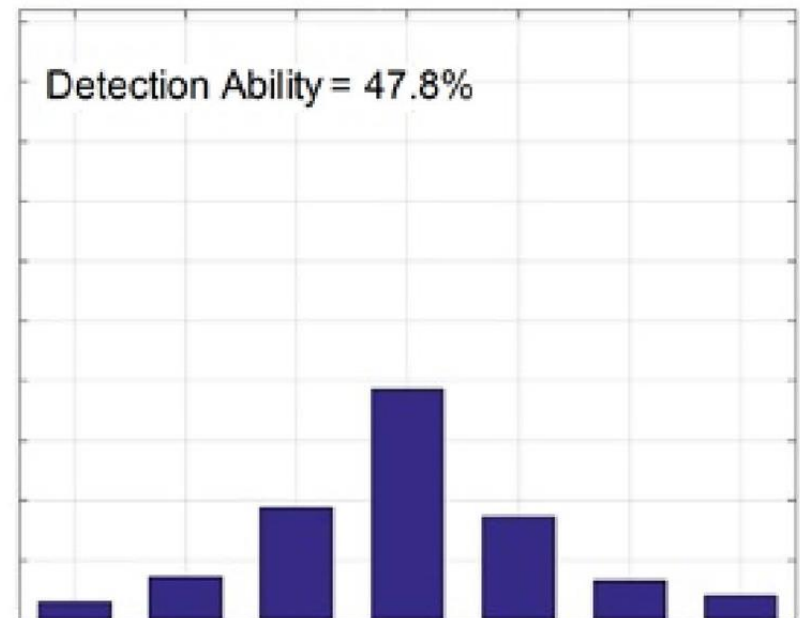
Offset detection power

Univariate analysis: **EAST**

White noise only



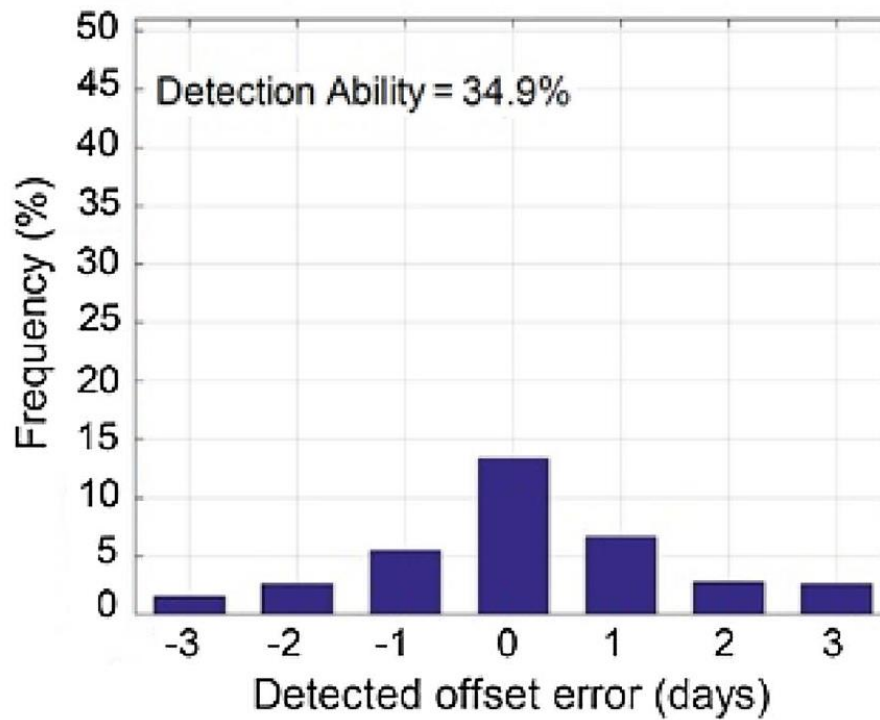
White noise + Flicker noise



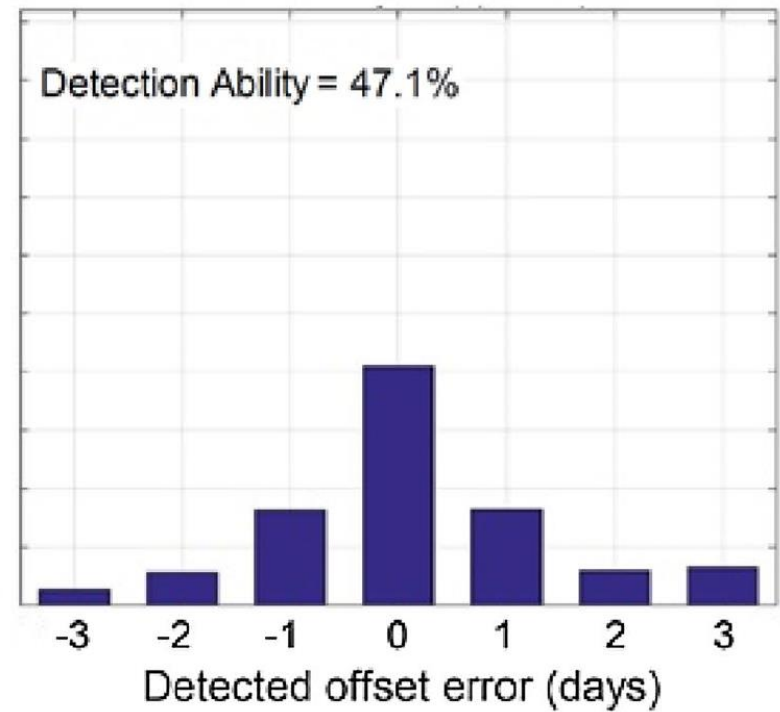
Offset detection power

Univariate analysis: **UP**

White noise only



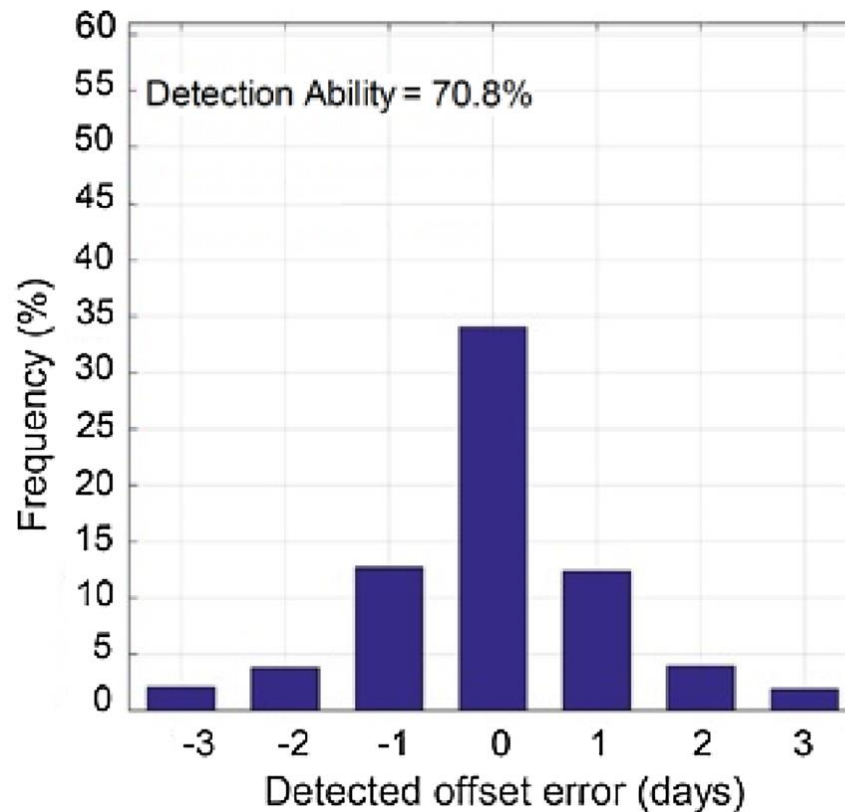
White noise + Flicker noise



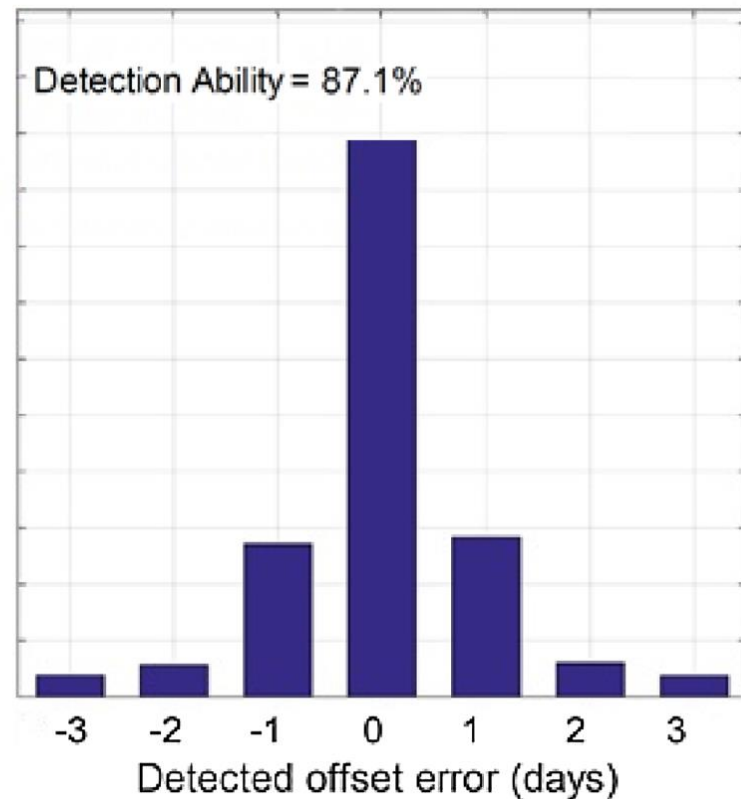
Offset detection power

Multivariate analysis: **N+E+U**

White noise only



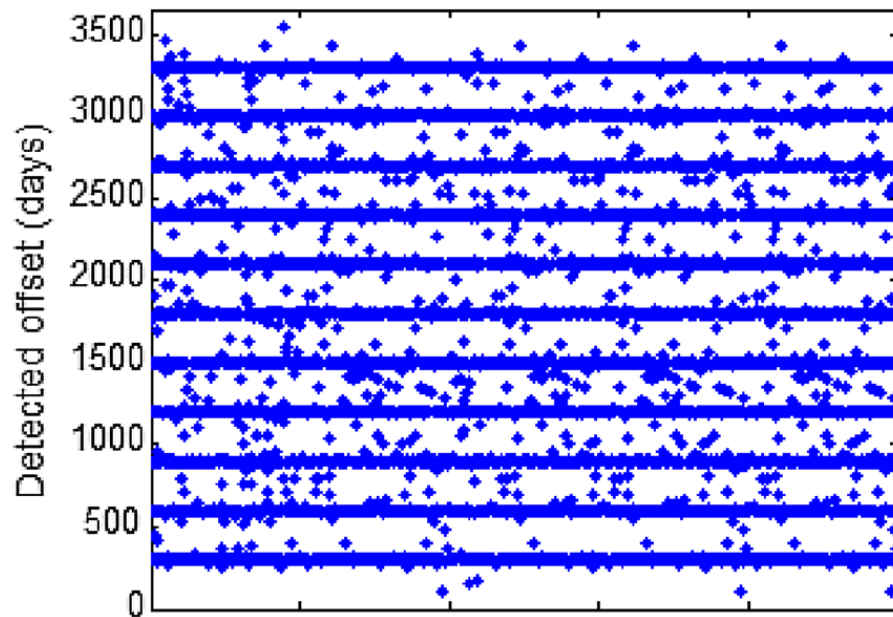
White noise + Flicker noise



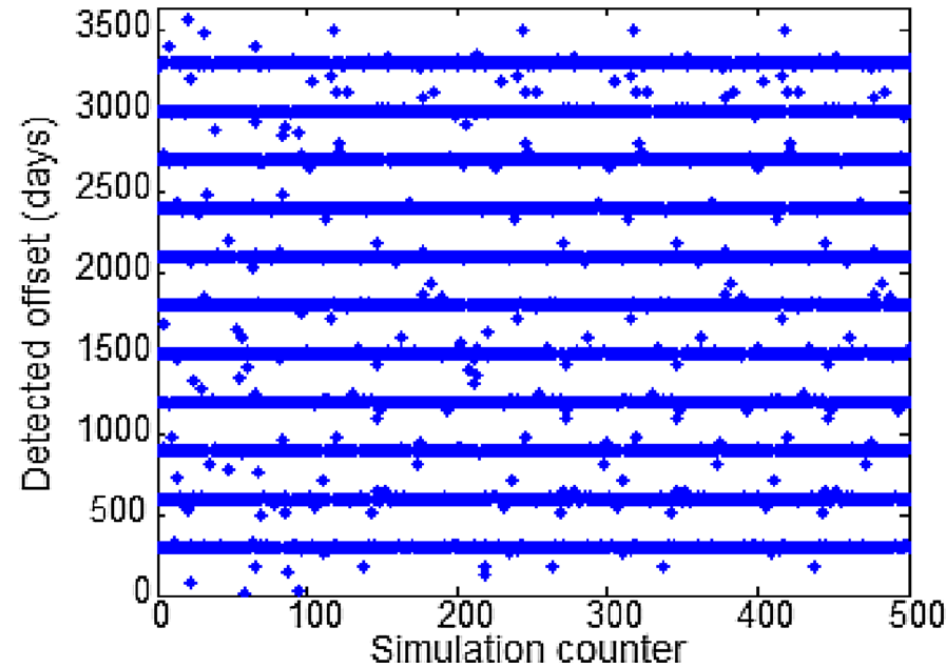
Offset detection power

Multivariate analysis: **N+E+U**

White noise only

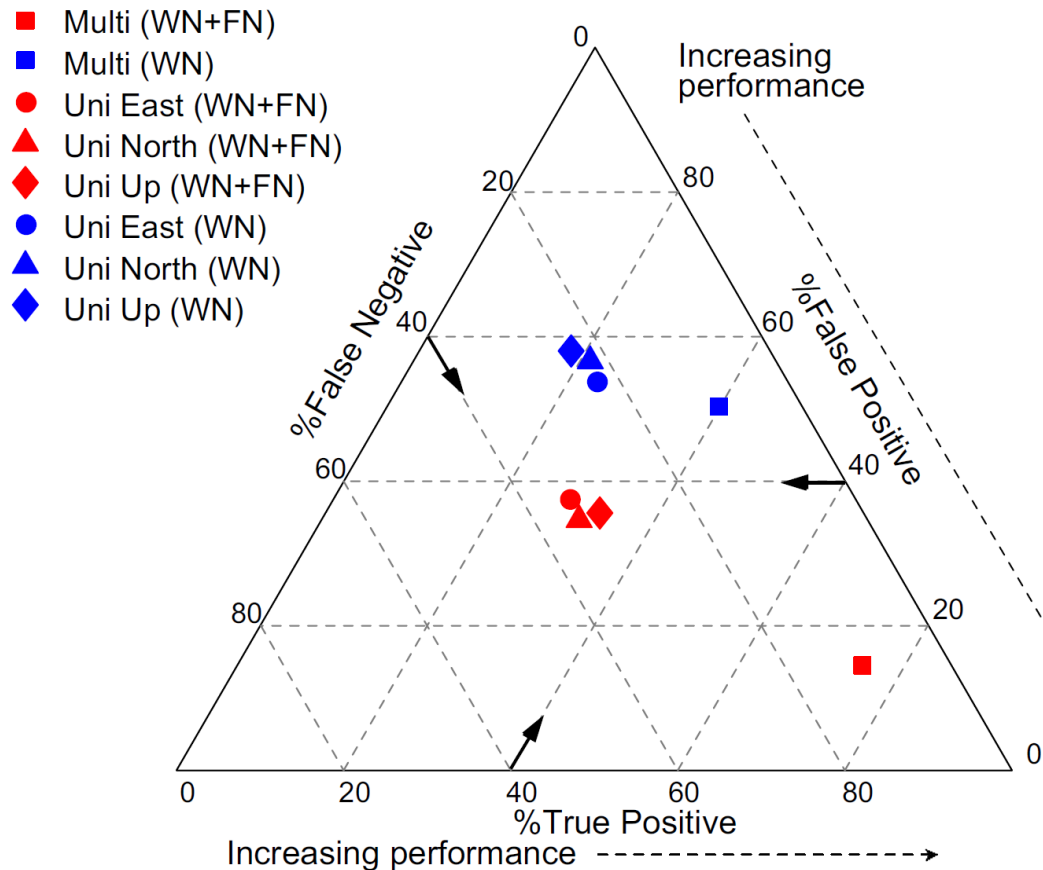


White noise + Flicker noise



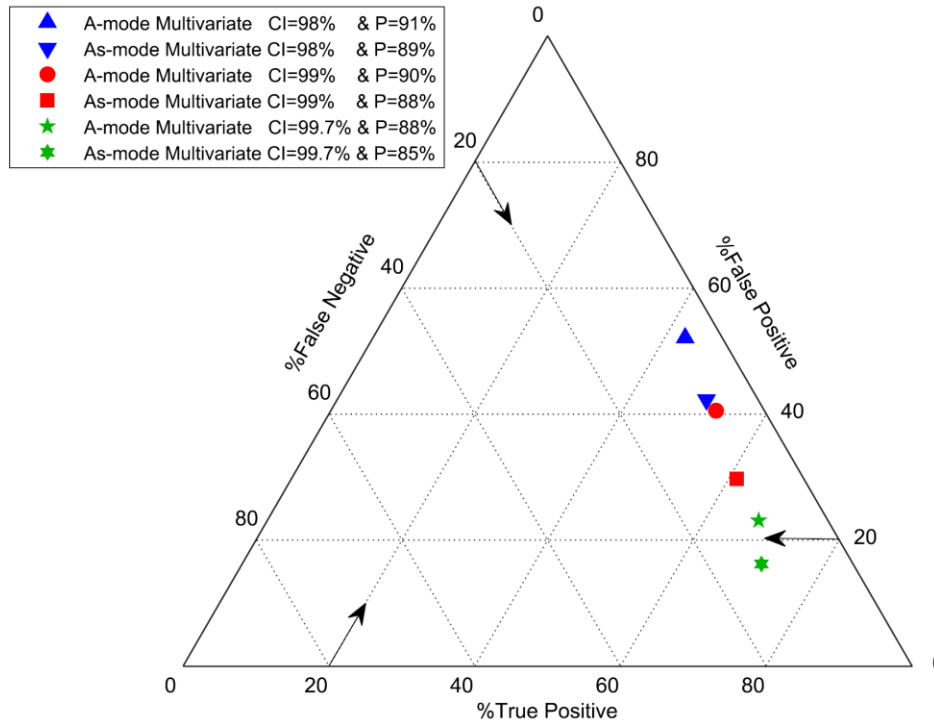
Offset detection power

Ternary plot: **Four cases**



Offset detection power

Ternary plot: Three multivariate cases



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Geophys. J. Int. (2020) **224**, 257–270
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GJI Gravity, Geodesy and Tides

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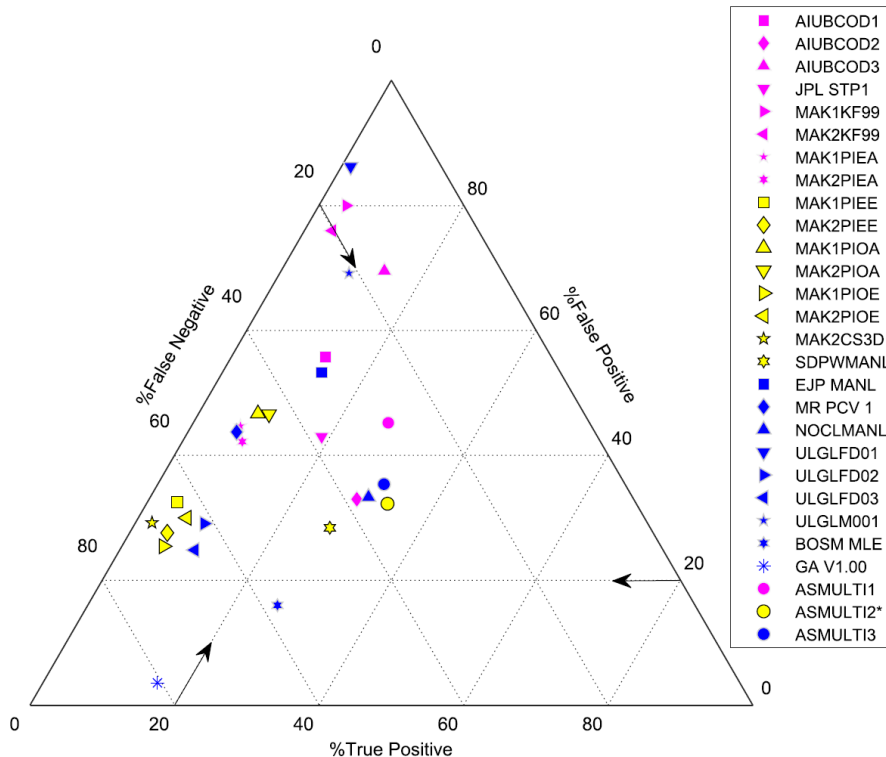
Improving offset detection algorithm of GNSS position time-series using spline function theory

S.M. Khazraei[✉] and A.R. Amiri-Simkooei[✉]



Offset detection power

Ternary plot: Four cases



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